

IMO 2021-22 Mathematics Training Exercise

Part 1

1. Suppose $a_1, a_2, a_3, \dots$ is a list of natural numbers satisfying $\sum_{k=1}^n a_k^3 = \left(\sum_{k=1}^n a_k \right)^2$.

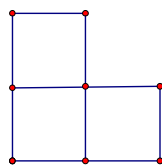
Then $a_n = n$, for all n .

2. Show that for $n \geq 2$, $\frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 1$.

3. Among $2n$ points in space, $n > 1$, and supposed they are connected by at least $n^2 + 1$ segments, then there exists a triangle.

4. For $n > 1$, $\frac{2^n - 1}{n}$ is never an integer.

5. An 1×1 square is removed from a $2^n \times 2^n$ block. Show that the block may be covered by triminos of the form



6. There are $2n-1$ satellites in space, the distances between satellites are all different. Each satellite is observing the satellite closest to him. Show that there exists at least one satellite who is not being observed by any other satellite.

7. (a) Show that a square may be dissected into n squares for $n \geq 6$.

- (b) Show that an equilateral triangle can be dissected into n isosceles triangles for $n \geq 6$.

8. Let m, n be positive integers and such that $\frac{m}{n} < 1$. Show that $\frac{m}{n}$ can be represented as reciprocals of *distinct* positive integers.

Part 2

1. $N(N+1)$ people of distinct heights are in a row. Show that one can remove $N(N-1)$ of them so that among the remaining $2N$ people, the two tallest one stick together, the third and fourth tallest people stick together, etc.
2. For any positive integer $n \geq 1$, the sequence $2, 2^2, 2^{2^2}, 2^{2^{2^2}}, \dots \pmod{n}$ is eventually constant.
3. An integer n is *good* if $n = a_1 + a_2 + \dots + a_k$, where $a_1, a_2, \dots, a_k$ are positive integers (not necessarily distinct) satisfying $\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} = 1$. It is known that integers from 33 to 73 are good, show that every integer ≥ 33 is good.
4. For $n \geq 3$, 2^n may be expressed in the form $7x^2 + y^2$, with x, y odd.
5. Show that $\sum_{r=0}^n (-1)^r \binom{n}{r} (n-r)^n = n!$.
6. (a) Let p be a prime number, $p \geq 3$. Show that for any $2p-1$ integers, there exist p of them, whose sum is divisible by p .
(b) Show that for any $2n-1$ integers, there exist n of them, whose sum is divisible by n .
7. Given a sequence of length n , $a_1, a_2, \dots, a_n$. If there is a part of the sequence of the form $a_k, a_{k+1}, \dots, a_{k+q-1}$, and whose (arithmetical) mean exceed 2017, then it is called a *dragon* and a_k is its (dragon) *head*. (In case $a_k > 2017$, then it itself is a dragon and a_k its own head.) Given that the original sequence has at least one dragon, show that the mean of all its head also exceed 2017.
8. Let $A = \{a_1, a_2, \dots, a_n\}$, with $a_1 < a_2 < \dots < a_n$. φ is a one-one map from A to A and such that $a_1 + \varphi(a_1) < a_2 + \varphi(a_2) < \dots < a_n + \varphi(a_n)$. Show that $\varphi(a_i) = a_i$ for all i .
9. There are n gas stations $A_1, A_2, \dots, A_n$ on a circle road. On station A_i a car gets a_i amount of oil. And from A_i to A_{i+1} , b_i amount of oil is spent ($A_{n+1} = A_1$). It is known $\sum a_i = \sum b_i$. Show that the car may start from a station, putting up oil, and move on, until it come a full circle.

10. Determine if there exists n with exactly 2017 prime divisors and such that $n \mid 2^n + 1$.

Part 3

1. An ordered pair (x, y) of integers is an *irreducible lattice point* if x and y are relatively prime. For any finite set S , show that there is a homogeneous polynomial in two variables, $f(x, y)$ with integer coefficients, of degree at least 1 such that $f(x, y) = 1$ for each (x, y) in the set S .
2. Let $a_1, a_2, \dots, a_n$ be distinct positive integers and let M be a set of $n-1$ positive integers not containing $s = a_1 + a_2 + \dots + a_n$. A grasshopper is to jump along the real axis, starting at the point 0 and making n jumps to the right with length $a_1, a_2, \dots, a_n$ in some order. Prove that the order can be taken in such a way that the grasshopper never lands on any point in M .
3. Let $P = A_1 A_2 \dots A_k$ be a convex polygon in the plane. The vertices $A_1, A_2, \dots, A_k$ have integral coordinates and lie on a circle. Let S be the area of P . An odd integer n is given such that the squares of the side lengths of P are integers divisible by n . Prove that $2S$ is an integer divisible by n .
4. σ is a *permutation* of $\{1, 2, \dots, n\}$. If $\sigma(i) > \sigma(j)$ for some $i < j$, then we have an *inversion*. The number of permutations of σ is s . Also define

$$d(\sigma) = \sum_{i=1}^n |\sigma(i) - i|.$$

Show that $s \leq d(\sigma)$.

5. A positive integer is *alternative* if its decimal digits are alternatively odd and even. Find all positive integers n such that n has an alternative multiple.

22 December 2021

Easy

1. Suppose $a_1, a_2, a_3, \dots$ is a list of natural numbers satisfying $\sum_{k=1}^n a_k^3 = \left(\sum_{k=1}^n a_k\right)^2$.

Then $a_k = k$.

True if $n=1$, since $a_1^3 = a_1^2$ and $a_1 \neq 0$ will imply $a_1 = 1$. Assume now the statement is true for natural numbers $1, 2, \dots, n$. By induction hypothesis,

$$1^3 + 2^3 + \dots + n^3 + a_{n+1}^3 = (1 + 2 + \dots + n + a_{n+1})^2, \text{ or}$$

$$\left(\frac{n(n+1)}{2}\right)^2 + a_{n+1}^3 = \left(\frac{n(n+1)}{2}\right)^2 + n(n+1)a_{n+1} + a_{n+1}^2,$$

simplify to get $(a_{n+1} - n - 1)(a_{n+1} + n) = 0$ and the result.

2. Show that for $n \geq 2$, $\frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 1$.

Indeed $\frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} < 1 - \frac{1}{n}$. One may proceed by induction.

(Of course one can show this by using the formula $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$.)

3. Among $2n$ points in space, $n > 1$, and supposed they are connected by at least $n^2 + 1$ segments, then there exists a triangle.

(Putnam 1956, Turan's problem) For $n=2$, they are connected by at least 5 segments, so at most 1 segment is missing, say AB , then we may simply choose a triangle formed by other points. By induction, assume if $2(n-1)$ points are connected by at least $(n-1)^2 + 1$ segments, then there exists a triangle. Now given $2n$ points connected by $n^2 + 1$ segments. Now consider a segment PQ , connecting P and Q , if at least one point is connected to both P and Q , then we have a triangle. If not, all other $2(n-1)$ points are connected to at most one of P or Q , but not both, hence that accounts for at most $2(n-1)$ segments, plus the segment PQ , there are at least $(n^2 + 1) - 2(n-1) - 1 = (n-1)^2 + 1$ segments left for the $2(n-1)$ points and we are done. (Note n^2 segments for $2n$ points is not good enough, consider the case n points are connected to other n points only.)

4. For $n > 1$, $\frac{2^n - 1}{n}$ is never an integer.

Suppose not, choose m the smallest integer with such property. Thus $m \mid 2^m - 1$.

Likewise $m \mid 2^{\phi(m)} - 1$. Take $d = \gcd(m, \phi(m))$, then $m \mid 2^d - 1$. But $m > 1$, thus

$d > 1$, and we have $d \mid m$ and $d \mid 2^d - 1$. Yet $1 < d \leq \phi(m) < m$, contradicting the

minimality of m . (This is induction in reverse, called the method of *infinite descent*.)

One may prove the statement like this, suppose n divides $2^n - 1$, take p the smallest

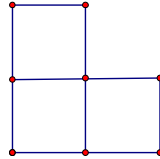
prime factor of n , then $p \geq 3$. Now $p \mid 2^{p-1} - 1$ and $p \mid 2^n - 1$, thus

$p \mid 2^{\gcd(n, p-1)} - 1 = 2^1 - 1 = 1$, a contradiction! Or using the order, let $n > 1$ be the

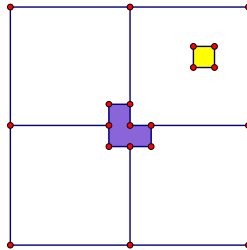
smallest integer satisfying $n \mid 2^n - 1$, let the order of $2 \bmod n$ be d , then $1 < d < n$.

And $d \mid n$, hence $2^d \equiv 1 \pmod{d}$, contradicting the minimality of n .)

5. An 1×1 square is removed from a $2^n \times 2^n$ block. Show that the block may be covered by triminos of the form



(Classical) Place the trimino as shown and apply the induction hypothesis.



6. There are $2n-1$ satellites in space, the distances between satellites are all different. Each satellite is observing the satellite closest to him. Show that there exists

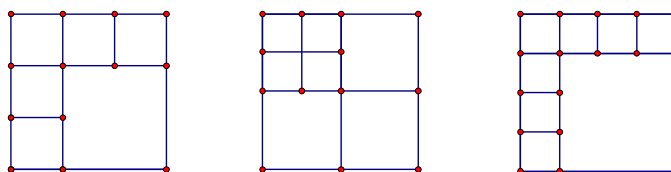
at least one satellite who is not being observed by any other satellite.

(Classical) For $n = 1$, it is clear. Now suppose there are $2n - 1$ satellites with $n \geq 2$. Now suppose A and B are two satellites whose distance is shortest among all other distance. Then A is observing B and vice versa. For the other $2n - 3$ satellites, by induction, there exists one satellite which is not being observed. (Minimal principle, also the statement is not true for $2n$, just pair up the satellites observing each other.)

7. (a) Show that a square may be dissected into n squares for $n \geq 6$.

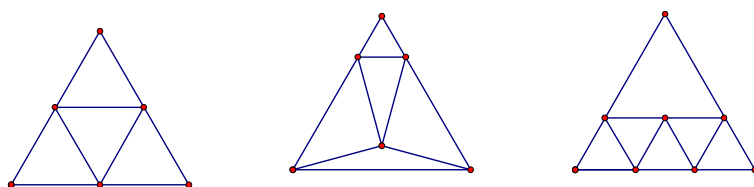
(b) Show that an equilateral triangle can be dissected into n isosceles triangles for $n \geq 6$.

(Classical) (a) A square may be dissected into 6, 7, or 8 squares as shown.



Apply the induction hypothesis from n to $n + 3$. Note a square cannot be dissected into 2, 3, or 5 squares.

(b) An equilateral triangle may be dissected into 4, 5 or 6 isosceles triangles as shown.



The waists of the middle triangle are right-angled triangles. Apply the induction hypothesis from n to $n + 3$ again.

8. Let m, n be positive integers and such that $\frac{m}{n} < 1$. Show that $\frac{m}{n}$ can be represented as reciprocals of *distinct* positive integers.

For $m = 1$, it is clear. Assume now the statement is true for all positive integers $< m$.

Now $n = qm - r$, where $0 \leq r < m < n$. Hence $\frac{m}{n} = \frac{1}{q} + \frac{r}{nq}$. By induction

$$\frac{r}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \dots + \frac{1}{n_k}, \text{ where } n_1 < n_2 < \dots < n_k. \text{ Thus}$$

$$\frac{m}{n} = \frac{1}{q} + \frac{1}{qn_1} + \frac{1}{qn_2} + \dots + \frac{1}{qn_k}.$$

Note $n_1 > 1$.

Medium

1. $N(N+1)$ people of distinct heights are in a row. Show that one can remove $N(N-1)$ of them so that among the remaining $2N$ people, the two tallest one stick together, the third and fourth tallest people stick together, etc.

(IMO2017) First the lemma. Assume we have N disjoint group of at least $N+1$ people in each, all have distinct heights. Then one can choose two people from each group so that among the chosen people, the two tallest ones go together, so are the third and the fourth tallest ones, and so on. The proof is by induction on N . For $N=1$, it is clear. Consider now the N groups $G_1, G_2, \dots, G_N$ with at least $N+1$ persons in each group for $N \geq 2$. Now let the people be enumerated by height $1, 2, \dots, N(N+1)$, from shortest to tallest, say. Find the *least* s such that two people among $1, 2, \dots, s$ are in the same group, say G_N . By minimality of s , the two people are s and i for some $i < s$. Now choose i and s , then forget about the group G_N , then remove $1, 2, \dots, s$ from other groups. By minimality of s , each of the remaining groups $G'_1, G'_2, \dots, G'_{N-1}$ has at least N elements, and we can apply the induction hypothesis. Together with i and s , we are done.

2. For any positive integer $n \geq 1$, the sequence $2, 2^2, 2^{2^2}, 2^{2^{2^2}}, \dots \pmod{n}$ is eventually constant.

(USAMO1991) In case of $n=1$, it is clear. Assume now the statement is true for all integers less than n . Assume now n is even, then $n = 2^k q$, with $k \geq 1$ and q is odd.

Let the original terms be $a_1, a_2, \dots (a_1 = 2, a_{i+1} = 2^{a_i}, i \geq 1)$. From the induction hypothesis, $a_1, a_2, a_3, \dots$ is eventually constant mod q . Also clearly for large enough

i , $a_i \equiv 0 \pmod{2^k}$. But q and 2^k are relatively prime, $2^k \mid a_{i+1} - a_i$ and $q \mid a_{i+1} - a_i$

will imply $n \mid a_{i+1} - a_i$, done. If n is odd, there exist positive integer $r < n$ such that

$2^r \equiv 1 \pmod{n}$, say we may take $r = \phi(n)$ the Euler totient function. Hence by induction $a_1, a_2, \dots$ is eventually constant mod r , say $a_i \equiv c \pmod{r}$. Thus

$a_{i+1} = 2^{a_i} = 2^{m_i r + c} = 2^{m_i r} 2^c \equiv 2^c \pmod{n}$. This implies $a_1, a_2, \dots$ is eventually constant mod n .

3. An integer n is *good* if $n = a_1 + a_2 + \dots + a_k$, where $a_1, a_2, \dots, a_k$ are positive integers (not necessarily distinct) satisfying $\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} = 1$. It is known that integers from 33 to 73 are good, show that every integer ≥ 33 is good.

(USAMO1978) We can show that if n is good, then $2n+8$ and $2n+9$ is good.

Indeed suppose $a_1, a_2, \dots, a_n$ is a partition of n , then $\frac{1}{2a_1} + \frac{1}{2a_2} + \dots + \frac{1}{2a_k} = \frac{1}{2}$. Since

$\frac{1}{2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{3} + \frac{1}{6}$, hence $(4, 4, 2a_1, 2a_2, \dots, 2a_k)$ and $(3, 6, 2a_1, 2a_2, \dots, 2a_k)$ are

partitions of $2n+8$ and $2n+9$ respectively. Hence from 33 we get 74 and 75. Denote by S_n the statement $n, n+1, \dots, 2n+7$ are good integers. Hence S_{33} is valid (given). From the last analysis, we get S_{34} is valid, and hence by induction S_n is good for $n \geq 33$.

4. For $n \geq 3$, 2^n may be expressed in the form $7x^2 + y^2$, with x, y odd.

(Euler) First $2^3 = 7 \cdot 1^2 + 1^2$.

Now suppose $2^n = 7x^2 + y^2$, with x, y odd. Now $2 = 7 \cdot \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2$. Thus

$$\begin{aligned} 2^{n+1} &= (7x^2 + y^2) \left(7 \cdot \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \right) \\ &= 7 \left(\frac{x}{2} + \frac{y}{2} \right)^2 + \left(\frac{7x}{2} - \frac{y}{2} \right)^2 \\ &= 7 \left(\frac{x}{2} - \frac{y}{2} \right)^2 + \left(\frac{7x}{2} + \frac{y}{2} \right)^2. \end{aligned}$$

Now x, y odd, hence $\frac{x+y}{2}, \frac{7x-y}{2}$ are integers. If $\frac{x+y}{2}$ odd, then

$\frac{7x-y}{2} = 4x - \frac{x+y}{2}$ odd. If $\frac{x+y}{2}$ even, then $\frac{x-y}{2} = \frac{x+y}{2} - y$ odd, so is $\frac{7x+y}{2} = 4x - \frac{x-y}{2}$ odd. Together we can proceed by induction. (Note $7x^2 + y^2$ is a norm and one can do the composition.)

5. Show that $\sum_{r=0}^n (-1)^r \binom{n}{r} (n-r)^n = n!$.

Generalize the equality to an equality of real variable $\sum_{r=0}^n (-1)^r \binom{n}{r} (x-r)^n = n!$.

Check a few cases. Now the equality is valid for $n=1$. Assume it is valid for n . Then

$$\begin{aligned}
& \sum_{r=0}^{n+1} (-1)^r \binom{n+1}{r} (x-r)^{n+1} \\
&= \sum_{r=0}^{n+1} (-1)^r \binom{n+1}{r} (x-r)^n (x-n-1+n+1-r) \\
&= \sum_{r=0}^{n+1} (-1)^r \binom{n+1}{r} (x-r)^n (x-n-1) + \sum_{r=0}^{n+1} (-1)^r \binom{n+1}{r} (x-r)^n (n+1-r) \\
&= (x-n-1) \sum_{r=0}^{n+1} (-1)^r \left(\binom{n}{r} + \binom{n}{r-1} \right) (x-r)^n + (n+1) \sum_{r=0}^n (-1)^r \binom{n}{r} (x-r)^n \\
&= (x-n-1) \left(\sum_{r=0}^n (-1)^r \binom{n}{r} (x-r)^n + \sum_{r=1}^{n+1} (-1)^r \binom{n}{r-1} (x-r)^n \right) + (n+1)! \\
&= (x-n-1) \left(n! + \sum_{r=0}^n (-1)^{r+1} \binom{n}{r} (x-1-r)^n \right) + (n+1)! \\
&= (x-n-1)(n! - n!) + (n+1)! \\
&= (n+1)!.
\end{aligned}$$

Note: $\binom{n}{r} = 0$ when $r < 0$ or $n > r$. Also $(n+1-r)$ cancels out with $\binom{n+1}{r}$.

6. (a) Let p be a prime number, $p \geq 3$. Show that for any $2p-1$ integers, there exist p of them, whose sum is divisible by p .

(b) Show that for any $2n-1$ integers, there exist n of them, whose sum is divisible by n .

(a) First note if p is prime, then $\binom{n}{p} \equiv \left\lfloor \frac{n}{p} \right\rfloor \pmod{p}$. Indeed among $0, 1, 2, \dots, p-1$,

one of them is congruent to $n \pmod p$, say $n = kp + i$, then $\frac{n-i}{p} = k = \left\lfloor \frac{n}{p} \right\rfloor$. Also $n(n-1)\dots(n-i+1)(n-i)\dots(n-p+1) \equiv i(i-1)\dots(p-1)\dots(i+1) \equiv (p-1)! \pmod p$. Hence $(p-1)! \binom{n}{p} \equiv (p-1)! \frac{n-i}{p} \pmod p$, cancelling out to get the result. Using this, get $\binom{2p-1}{p} \equiv \left\lfloor \frac{2p-1}{p} \right\rfloor = 1 \pmod p$.

Now suppose none of the sum $a_{i_1}, a_{i_2}, \dots, a_{i_p}$ is divisible by p . Then by the Fermat's little theorem, $(a_{i_1} + a_{i_2} + \dots + a_{i_p})^{p-1} \equiv 1 \pmod p$. There are $\binom{2p-1}{p}$ of these sums. Summing up all these sums, $\sum (a_{i_1} + a_{i_2} + \dots + a_{i_p})^{p-1} \equiv \binom{2p-1}{p} \equiv 1 \pmod p$. However when we expand $(a_{i_1} + a_{i_2} + \dots + a_{i_p})^{p-1}$, we get terms such as $a_{i_1}^{\alpha_1} a_{i_2}^{\alpha_2} \dots a_{i_l}^{\alpha_l}$, where $l \leq p-1$. The number of the (same) corresponding term is $\binom{2p-1-l}{p-l}$, i.e. choosing $p-l$ terms other than $a_{i_1}, a_{i_2}, \dots, a_{i_l}$. Note

$$\binom{2p-1-l}{p-l} = \frac{(2p-1-l)\dots(p+1)p}{(p-l)!} \text{ divisible by } p.$$

(b) The statement is valid when n is a prime p . Now we proceed by induction, assume the statement if n has less factors s , and assume n has s ($s > 2$) factors, say $n = ab$.

By induction, we can choose $m_{1,1}, m_{1,2}, \dots, m_{1,a}$, whose sum is divisible by a . From

the remaining $2n-1-a$ numbers, can choose $m_{2,1}, m_{2,2}, \dots, m_{2,a}$, whose sum is divisible by a . Continuing in this matter, until the last $2n-1-(2b-2)a = 2a-1$ numbers $m_{2b-1,1}, m_{2b-1,2}, \dots, m_{2b-1,a}$, whose sum is divisible by a . The quotients of the sums divisible by a are $m_1, m_2, \dots, m_{2b-1}$. From these quotients, we can choose b of them, say $m_1, m_2, \dots, m_b$ whose sum is divisible by b , say $m_1 + m_2 + \dots + m_b = kb$. Hence members of these groups

$$((m_{1,1} + m_{1,2} + \dots + m_{1,a}) + \dots + (m_{b,1} + m_{b,2} + \dots + m_{b,a})) = a(m_1 + m_2 + \dots + m_b) = kab$$

as required.

7. Given a sequence of length n , $a_1, a_2, \dots, a_n$. If there is a part of the sequence of the

form $a_k, a_{k+1}, \dots, a_{k+q-1}$, and whose (arithmetical) mean exceed 2017, then it is called a *dragon* and a_k is its (dragon) *head*. (In case $a_k > 2017$, then it itself is a dragon and a_k its own head.) Given that the original sequence has at least one dragon, show that the mean of all its head also exceed 2017.

(Chinese MO1988) We prove by induction using the length of the sequence. Clearly the statement is true if $n = 1$. Now we assume the statement is true for all sequences of length $< n$.

Method 1. Consider the sequence $a_1, a_2, \dots, a_n$. If a_1 is not the head of any dragon, then it has the same number of heads as $a_2, \dots, a_n$. By induction the statement is valid and hence true for the entire sequence. If a_1 is the head of a dragon, say $a_1, a_2, \dots, a_m$, then by definition its mean exceed 2017. Removing those elements which are not heads in the *original sequence*, each of these elements has value ≤ 2017 , the remaining elements still have mean > 2017 . For the remaining part $a_{m+1}, a_{m+2}, \dots, a_n$, the heads are same as in the original sequence, hence by induction, its mean exceed 2017. (This sequence may be empty.) Together we see the mean of all heads exceed 2017.

Method 2. Consider the sequence $a_1, a_2, \dots, a_n$. If each term is a head of a certain dragon, then we can partition the sequence of disjoint dragons, and the mean of all heads will exceed 2017. Otherwise consider x which is the first term who is not the head of any dragon, then $x < 2017$, and the removal of x will not change the fact that all terms before x can serve as the heads of dragons. We now have a sequence of $n - 1$ terms and proceed by induction.

8. Let $A = \{a_1, a_2, \dots, a_n\}$, with $a_1 < a_2 < \dots < a_n$. φ is a one-one map from A to A and such that $a_1 + \varphi(a_1) < a_2 + \varphi(a_2) < \dots < a_n + \varphi(a_n)$. Show that $\varphi(a_i) = a_i$ for all i .

It is clear for $n = 1$. Assume the statement is true for all sets of length $< n$. Now suppose $\varphi(a_k) = a_1$. Then $a_1 + \varphi(a_1) < a_2 + \varphi(a_2) < \dots < a_k + \varphi(a_k) = a_1 + a_k$. Hence all $\varphi(a_1), \varphi(a_2), \dots, \varphi(a_{k-1})$ smaller than a_k . Then φ is a one-one map from $\{a_1, a_2, \dots, a_k\}$ to $\{a_1, a_2, \dots, a_k\}$. If $k = n$, then $a_1 + a_n$ is larger than every $a_i + \varphi(a_i)$, $1 \leq i \leq n - 1$. But among these terms there is one $\varphi(a_i) = a_n$, contradiction. Hence $k < n$. By induction $\varphi(a_i) = a_i$, for all $1 \leq i \leq k$. Apply induction to the other part we have $\varphi(a_i) = a_i$, for all $k + 1 \leq i \leq n$.

9. There are n gas stations $A_1, A_2, \dots, A_n$ on a circle road. On station A_i a car gets a_i amount of oil. And from A_i to A_{i+1} , b_i amount of oil is spent ($A_{n+1} = A_1$). It is known $\sum a_i = \sum b_i$. Show that the car may start from a station, putting up oil, and move on, until it come a full circle.

(Classical). Choose a station with $a_i \geq b_i$, say $a_n \geq b_n$. Treat A_n and A_1 as one station A_1' . Store oil reserve $a_1 + a_n - b_n$. From A_{n-1} to A_1' , b_{n-1} oil will be spent, from A_1' to A_2 , b_2 spent. By induction, there exists a station so that a car may start to run through the whole circle. (If that is A_1' , just start from A_n).

10. Determine if there exists n with exactly 2017 prime divisors and such that $n \mid 2^n + 1$.

(IMO2000) By induction on k , $k \geq 1$, the following proposition: for any k , there exists a positive integer $n = n(k)$ such that $3 \mid n$, n has exactly k distinct prime divisors and $n \mid 2^n + 1$. For $k = 1$, we have $n(1) = 3$ and $3 \mid 2^3 + 1$. Assume now the statement for k and $n = n(k)$. Since $n \mid 2^n + 1$, n is odd. Now $2^{3n} + 1 = (2^n + 1)(2^{2n} - 2^n + 1)$. We claim $2^{2n} - 2^n + 1$ has a prime factor p such that $p \nmid 2^n + 1$. It follows that the number $n(k+1) = 3np$ has $k+1$ distinct prime divisors and $3np \mid 2^{3np} + 1 = (2^{3n} + 1)(2^{3n(p-1)} - 3^{3n(p-2)} + \dots - 2^{3n} + 1)$. We are done. (Proof of the claim: $a^3 + 1 = (a+1)(a^2 - a + 1)$. Assume any p divides both $(a^2 - a + 1)$ and $a + 1$. Now $(a^2 - a + 1) = (a+1)(a-2) + 3$, we obtain $p \mid 3$. Then $(a^2 - a + 1) = p^l$. Since $3 \mid a+1$ and $3 \mid (a-2) = (a+1) - 3$, it follows that $a^2 - a + 1 \equiv 3 \pmod{9}$. Together we got $a^2 - a + 1 = 3$, a contradiction. (Indeed we show $3^{n+1} \parallel 2^n + 1$. Using the fact $2^{3n} + 1$ produces a new prime divisor p_i every time, we just choose $3^k p_1 p_2 \dots p_{2016}$ and k large enough.)

(Inventiveness in searching for proper inductive procedure.)

Hard

1. An ordered pair (x, y) of integers is an *irreducible lattice point* if x and y are relatively prime. For any finite set S , show that there is a homogeneous polynomial in two variables, $f(x, y)$ with integer coefficients, of degree at least 1 such that $f(x, y) = 1$ for each (x, y) in the set S .

(IMO2017) Assume the irreducible lattice points are $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, we may assume no two are collinear from the origin. Now we proceed by induction. For $n=1$, since x_1 and y_1 are relatively prime, we have $cx_1 + dy_1 = 1$. Thus choosing $f(x, y) = cx + dy$ will do. For $n \geq 2$, we have a homogeneous polynomial $g(x, y)$ with $g(x_1, y_1) = g(x_2, y_2) = \dots = g(x_n, y_n) = 1$. Assume $j = \deg g$. Define now $g_n(x, y) = \prod_{k=1}^{n-1} (y_k x - x_k y)$, and $a_n = g(x_n, y_n)$. By assumption $a_n \neq 0$. Take some integers c, d such that $cx_n + dy_n = 1$. Construct $f(x, y)$ in the form

$$f(x, y) = g(x, y)^K - C \cdot g_n(x, y) \cdot (cx + dy)^L,$$

where K and L are some positive integers and C is some integer, assume $L = Kj - n + 1$ to keep homogeneity. Note

$$g(x_1, y_1) = g(x_2, y_2) = \dots = g(x_{n-1}, y_{n-1}) = 1 \text{ and}$$

$$g_n(x_1, y_1) = g_n(x_2, y_2) = \dots = g_n(x_{n-1}, y_{n-1}) = 0, \text{ thus}$$

$$f(x_1, y_1) = f(x_2, y_2) = \dots = f(x_{n-1}, y_{n-1}) = 1.$$

$$\text{Note } f(x_n, y_n) = g(x_n, y_n)^K - C \cdot g_n(x_n, y_n) \cdot (cx_n + dy_n)^L = g(x_n, y_n)^K - Ca_n.$$

We now want to find K so that $g(x_n, y_n)^K \equiv 1 \pmod{a_n}$, then we can find C so that

$$f(x_n, y_n) = 1. \text{ This can be done using the Fermat's little theorem (details omitted).}$$

2. Let $a_1, a_2, \dots, a_n$ be distinct positive integers and let M be a set of $n-1$ positive integers not containing $s = a_1 + a_2 + \dots + a_n$. A grasshopper is to jump along the real axis, starting at the point 0 and making n jumps to the right with length $a_1, a_2, \dots, a_n$ in some order. Prove that the order can be taken in such a way that the grasshopper never lands on any point in M .

(IMO2009) We represent a route by $(i_1, i_2, \dots, i_n)$ if the grasshopper makes the jumps $a_{i_1}, a_{i_2}, \dots, a_{i_n}$ in that order. Now we proceed by induction. For $n=1, 2$, it is clear.

We now suppose the claim holds for all values smaller than n . We may assume $a_1 < a_2 < \dots < a_n$. Let $d = \min M$.

Case 1. $d < a_n$.

Case 1.1. $a_n \notin M$. Apply the induction hypothesis with a shift by $-a_n$, we obtain

that the grasshopper can jump from a_n to s not landing on the point of $M \setminus \{d\}$ using the jumps $a_1, a_2, \dots, a_{n-1}$. Putting a_n to the beginning of this sequence of jumps provides a desired result.

Case 1.2. $a_n \in M$. Consider the n pairwise disjoint sets

$\{a_n\}, \{a_1, a_1 + a_n\}, \{a_2, a_2 + a_n\}, \dots, \{a_{n-1}, a_{n-1} + a_n\}$. One of them does not intersect

M , say $\{a_i, a_i + a_n\}$. Then $|M \cap \{a_i + a_n, s\}| \leq n-3$ since $d, a_n < a_i + a_n$. Apply the induction hypothesis with a shift of $-(a_i + a_n)$, we derive that the grasshopper can jump from $a_i + a_n$ to s not landing on the points of M using all jumps except a_i and a_n . Putting a_i, a_n to the beginning of this sequence of jumps provides a desired route.

Case 2. $d \geq a_n$. Let $M' = M \setminus \{d\}$. By the induction hypothesis with a shift by

$-a_n$, the grasshopper can jump from a_n to s not landing on the points of M' using the jumps $a_1, a_2, \dots, a_{n-1}$. Let the corresponding indices be $(i_1, \dots, i_{n-1})$.

Case 2.1. The grasshopper does not step on d on this route (including the starting point). Then $d > a_n$. The sequence $U = \{n, i_1, i_2, \dots, i_{n-1}\}$ provides a desired route.

Case 2.2. The grasshopper steps on d on this route. Then d is the only point from M that the grasshopper meets on this route. There is some integer k such that

$0 \leq k < n-1$ and $a_n + \sum_{j=1}^k a_{i_j} = d$. Consider the route represented by

$(i_1, \dots, i_{k+1}, n, i_{k+2}, \dots, i_{n-1})$. Since $\sum_{j=1}^{k+1} a_{i_j} < \sum_{j=1}^k a_{i_j} + a_n = d$, the grasshopper will not

land on the points of M during the first $k+1$ jumps. During the remaining part of this route it lands on the same points as in the route U and these points are not less

than $\sum_{j=1}^{k+1} a_{i_j} + a_n > d$. Hence it will not land on points of M on this part as well, i.e.

on the whole route.

3. Let $P = A_1A_2\dots A_k$ be a convex polygon in the plane. The vertices $A_1, A_2, \dots, A_k$ have integral coordinates and lie on a circle. Let S be the area of P . An odd integer n is given such that the squares of the side lengths of P are integers divisible by n . Prove that $2S$ is an integer divisible by n .

(IMO2016) Using shoelace lemma (1.1) or Pick's formula (1.2), we know $2S$ is an integer. We may assume $n = p^t$, where p is an odd prime. Assume $P = A_1A_2\dots A_k$ and let $A_{k+i} = A_i$ for $i \geq 1$. Using Heron and Brahmagupta (2.3a, 2.4a), we immediately get the statement true for $k = 3, 4$. Now for $k \geq 5$, if we can find a diagonal divisible by n , then we may apply the induction procedure. So we assume none of the diagonals is divisible by n and try to get a contradiction.

We need a definition, denote by $v_p(r)$ the *exponent* of p in the factorization of r .

(Clearly the notation comes from the word *valuation* in algebra).

Now the key observation is to make the *crucial* claim

Claim. $v_p(A_1A_m^2) > v_p(A_1A_{m+1}^2)$ for $2 \leq m \leq k-1$.

Proof of the Claim. For $m = 2$, it is clear, because p^t is divisible by $A_1A_2^2$ (given condition) and $A_1A_3^2$ is not (assumption), namely $v_p(A_1A_2^2) > t > v_p(A_1A_3^2)$. Now we suppose $v_p(A_1A_2^2) > v_p(A_1A_3^2) > \dots > v_p(A_1A_m^2)$, where $3 \leq m \leq k-1$ is established.

Apply Ptolemy (2.1) to the cyclic quadrilateral $A_1A_{m-1}A_mA_{m+1}$, we get

$$A_1A_{m+1} \times A_{m-1}A_m + A_1A_{m-1} \times A_mA_{m+1} = A_1A_m \times A_{m-1}A_{m+1}.$$

Taking squares, get

$$\begin{aligned} A_1A_{m+1}^2 \times A_{m-1}A_m^2 &= A_1A_{m-1}^2 \times A_mA_{m+1}^2 + A_1A_m^2 \times A_{m-1}A_{m+1}^2 \\ &\quad - 2A_1A_{m-1} \times A_mA_{m+1} \times A_1A_m \times A_{m-1}A_{m+1}. \end{aligned} \quad (1)$$

From (5.1), we get $-2A_1A_{m-1} \times A_mA_{m+1} \times A_1A_m \times A_{m-1}A_{m+1}$ an integer. We now consider exponents in (5.1). Indeed $v_p(A_1A_{m-1}^2) > v_p(A_1A_m^2)$ (induction hypothesis). Also

$v_p(A_mA_{m+1}^2) \geq t > v_p(A_{m-1}A_{m+1}^2)$ (assumption). This gives

$$v_p(A_1A_{m-1}^2 \times A_mA_{m+1}^2) > v_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2). \quad (2)$$

Hence we have

$$\begin{aligned} v_p(4A_1A_{m-1}^2 \times A_mA_{m+1}^2 \times A_1A_m^2 \times A_{m-1}A_{m+1}^2) &= v_p(A_1A_{m-1}^2 \times A_mA_{m+1}^2) + v_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2) \\ &> 2v_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2) \end{aligned}$$

From (2). This implies

$$v_p(2A_1A_{m-1} \times A_mA_{m+1} \times A_1A_m \times A_{m-1}A_{m+1}) > v_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2). \quad (3)$$

Combine (1), (2), (3) to get

$$v_p(A_1A_{m+1}^2 \times A_{m-1}A_m^2) = v_p(A_1A_m^2 \times A_{m-1}A_{m+1}^2).$$

By $v_p(A_{m-1}A_m^2) \geq t > v_p(A_{m-1}A_{m+1}^2)$, get $v_p(A_1A_{m+1}^2) < v_p(A_1A_m^2)$. The claim then follows from induction.

Finally from the claim, $t > v_p(A_1A_3^2) > v_p(A_1A_4^2) > \dots > v_p(A_1A_k^2) \geq t$, a contradiction!

(Note: one may find a similar proof from the internet. Though the arguments a bit complicated, it is still the most natural one.)

4. σ is a *permutation* of $\{1, 2, \dots, n\}$. If $\sigma(i) > \sigma(j)$ for some $i < j$, then we have an *inversion*. The number of permutations of σ is s . Also define

$$d(\sigma) = \sum_{i=1}^n |\sigma(i) - i|.$$

Show that $s \leq d(\sigma)$.

Induction on s . Clear if $s = 0$. (Not too hard to see in case of $s = 1$.) Now we assume the statement is valid when the number of inversions $< s$. Assume now we have a permutation with s inversions. Clearly if $\sigma(1) = 1$, we can remove 1 because both s and $d(\sigma)$ are not affected. So we assume $\sigma(1) = t > 1$ (or remove first few numbers until $\sigma(i) > i$. Assume $\sigma(j) = 1$.

Define σ' another permutation such that $\sigma'(1) = 1, \sigma'(j) = t, \sigma'(i) = \sigma(i), i \neq 1, j$.

Schematically

$$\sigma: \begin{array}{cccccc} 1 & 2 & 3 & & j-1 & j \\ t & \sigma(2) & \sigma(2) & \cdots & \sigma(j-1) & 1 \end{array}.$$

$$\sigma': \begin{array}{cccccc} 1 & 2 & 3 & & j-1 & j \\ 1 & \sigma(2) & \sigma(2) & \cdots & \sigma(j-1) & t \end{array}.$$

Hence $d(\sigma) - d(\sigma') = (j-1) + (t-1) - |j-t| = \min(2t-2, 2j-2)$. On the other hand,

among $\sigma(2), \sigma(3), \dots, \sigma(j-1)$, g of them less than t , and $j-2-g$ larger than t . In σ , the first $j-1$ numbers form an inversion with 1, among $\sigma(2), \sigma(3), \dots, \sigma(j-1)$, g of them form inversions with 1. In σ' , among $\sigma(2), \sigma(3), \dots, \sigma(j-1)$, $j-2-g$ of them form inversions with t . So the difference

$$s - s' = (j-1) + g - (j-2-g) = 2g + 1.$$

Clearly $g \leq j-2$. But there are at most $t-2$ numbers less than t (except 1). Thus $g \leq t-2$. We get $0 \leq s - s' \leq \min(2j-3, 2t-3)$. From the above, we get $d(\sigma) - d(\sigma') \geq s - s' + 1$, or $d(\sigma) \geq s + d(\sigma') - s' + 1$. Since $s' < s$, we get by induction $s' < d(\sigma')$. And finally $d(\sigma) \geq s$.

5. A positive integer is *alternative* if its decimal digits are alternatively odd and even. Find all positive integers n such that n has an alternative multiple.

(IMO2004). n is alternative if and only if n is not divisible by 20.

22 December 2021

Important things to note

All about Induction

Induction variable

Strengthen the initial conditions and choose wisely

Set up the beginning step

Induction process Inventiveness